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$$\text{LHS} = (\tan A + \sin A)^2 + (\tan A - \sin A)^2$$

$$= \tan^2 A + \sin^2 A + 2 \tan A \sin A - (\tan^2 A + \sin^2 A - 2 \tan A \sin A)$$

$$= \tan^2 A + \sin^2 A + 2 \tan A \sin A - \tan^2 A - \sin^2 A + 2 \tan A \sin A$$

$$= 4 \tan A \sin A$$

Now solving RHS, we get,

$$= 4 \sqrt{mn}$$

$$= 4 \sqrt{(\tan A + \sin A)(\tan A - \sin A)}$$

$$= 4 \sqrt{\tan^2 A - \sin^2 A}$$

$$= 4 \sqrt{\frac{\sin^2 A}{\cos^2 A} - \sin^2 A} \quad \dots \quad \left[\tan^2 A = \frac{\sin^2 A}{\cos^2 A} \right]$$

$$= 4 \sqrt{\frac{\sin^2 A - \cos^2 A \sin^2 A}{\cos^2 A}}$$

$$= 4 \sqrt{\frac{\sin^2 A (1 - \cos^2 A)}{\cos^2 A}}$$

$$= 4 \sqrt{\frac{\sin^2 A \times \sin^2 A}{\cos^2 A}} \quad \dots \quad [\sin^2 A = 1 - \cos^2 A]$$

$$= \frac{4 \sin A \times \sin A}{\cos A}$$

$$= 4 \tan A \sin A$$

$\therefore \text{LHS} = \text{RHS}$, Hence proved

33 → Given - linear equations $x - y + 1 = 0$, $3x + 2y - 12 = 0$

In $x - y + 1 = 0$, when

x	1	2	5
y	2	3	6

In $3x + 2y - 12 = 0$

x	1	2
y	4.5	3